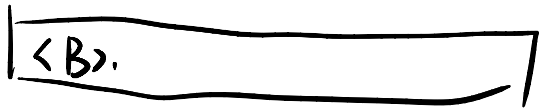
(lec 10)

a TM M that print $\langle M \rangle$ on the tape

$M : A \to B$.

A writes $\langle B \rangle$ to the tape



B writes $\langle A \rangle$ to the tape

and swap it with $\langle B \rangle$.

function $q : \{0,1\}^* \to \{0,1\}^*$, $q(w) = \langle M_w \rangle$

M_w = on input x

1. write w to the tape

✓ q is computable

B = on input w

1. compute $q(w)$
2. write $q(w)$ to the tape

   and swap$(q(w), w)$.

A = on any input

1. erase the input
2. write $\langle B \rangle$ to the tape

run A:

$\boxed{\langle B \rangle}$ $\longrightarrow$ run B. $\boxed{q(\langle B\rangle)\ \langle B\rangle}$

$q(\langle B\rangle)$ 就是$\langle A\rangle$!!

定理1: Recursion Theorem

For any TM T that takes an input $\langle M\rangle$ and x,

$\exists$ TM R, s.t. $R(x) = T(\langle R\rangle, x)$.

证:

T

R: $A \to B \to T$

A writes $\langle B\rangle\langle T\rangle$ to the tape

$\boxed{x\langle B\rangle\langle T\rangle}$

B writes $\langle A\rangle$ to the tape

and reorder.

$\boxed{\underbrace{\langle A\rangle\langle B\rangle\langle T\rangle}_{R}\ x}$

$T(\langle R\rangle, x)$.

#.

例:

可以用 Rec Theorem 证明 halt:

反证: 设 HALT 可计算:

$M =$ on input w

1. obtain $\langle M \rangle$.
2. compute HALT(M, w)
3. if HALT:

   loop

   else:

   halt

矛盾.

#.

例:

We say TM M 最小 if $\forall M' \Leftrightarrow M$, $|\langle M' \rangle| \geq |\langle M \rangle|$.

MINI (M) 不可计算.

证: 反证.

$M =$ on input x

1. obtain $\langle M \rangle$.
2. for s in $\{0,1\}^*$, 升序.

if MIN(S)==1 and $|S| > |\langle M\rangle|$.
run S on x
return $S(x)$

$M \Leftrightarrow S$ 但 $|\langle M\rangle| < |S|$

$\Downarrow$

S 非最小

矛盾.

#.

例: (fixed pointed) 定理.

let t 可计算. $\exists$ TM. F, s.t. $t(\langle F\rangle)=\langle M\rangle$,

且 $M \Leftrightarrow F$

证:

F = on input w

1. obtain $\langle F\rangle$.
2. compute $t(\langle F\rangle)$, let $\langle M\rangle = t(\langle F\rangle)$.
3. simulate M on w.

哥德尔不完备定理.

let $T \subseteq \{0,1\}^*$ be a set (set of true statement.

A proof system for T is a Turing Machine V that satifies:

(1). (effectiveness). $\forall x, t \in \{0,1\}^*$

$V(x,t)$ halt, return $0/1$

(2). (soundness) $\forall x \notin T, \forall t \in \{0,1\}^*$.

$V(x,t) = 0$

We say V 完备 (complete) if

$\forall x \in T, \exists t, V(x,t) = 1$

定理:

$\exists T$, 无完备证明系统

证:

T = set of all true statements of:

" a TM M halts on 0" or

"a TM M not halts on 0"

if T 有完备系统 V

并行做:
- run M on 0
- let x = "M does not halt on 0"
  for $t \in \{0,1\}^*$ 升序
    run $V(x,t)$.
    if $V(x,t) == 1$
      return 0

只有一个停机

#.

Quantified Integer Statement

符号
+ × =
∧ ∨ ¬
∀ ∃ (quantifier)

Given a QIS Φ

$$\mathrm{EVAL}(\phi) = \begin{cases} 1 & \text{if } \Phi \text{ is true} \\ 0 & \text{false.} \end{cases}$$

不可计算.

定理: let T be the set of QIS,

T 无完备 proof system.

Chapter 8 Running Time (lec 11)

定义: Running Time (TM).

Let $T: N \to N$. We say TM M Running Time $T(n)$, 若对于充分大的 n, $\forall x \in \{0,1\}^n$, M 在至多 $T(n)$ 步停机.

$\mathrm{TIME}_{TM}(T(n))$: {bool function F : F 可计算 with $T(n)$ Running Time

例: $\mathrm{TIME}_{TM}(10n^3) \subseteq \mathrm{TIME}_{TM}(2^n)$

定义: $P = \bigcup_{c=1,2,\cdots} \mathrm{TIME}_{TM}(n^c)$.

P C EXP

$P \subseteq EXP$

$$EXP = \bigcup_{c=1,2,\dots} TIME_{TM}(2^{n^c})$$

定义. Running Time (NAND-RAM program)

Let $T: N \to N$. running time of $T(n)$.

if $\dots n$. $\forall x \in \{0,1\}^n$, 在 $T(n)$ 行后停机.

$$TIME_{RAM}(T(n)) = \{F : F \text{ 可计算且 } T(n) \text{ 行内}\}$$

定理:

$$TIME_{TM}(T(n)) \subseteq TIME_{RAM}(10\,T(n)) \subseteq TIME_{TM}(T(n)^4)$$

$$P = \bigcup_{c=1,2,\dots} TIME_{RAM}(n^c)$$

$P \subsetneq EXP$

$$EXP = \bigcup_{c=1,2,\dots} TIME_{RAM}(2^{n^c})$$

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nice function:
$$\begin{cases} T(n) \geq n \\ T \text{ 单调不降} \\ T \text{ 本身在 } T(n) \text{ 可计算} \end{cases}$$

定理: TIME hierarchy theorem

$\forall$ nice function $T: N \to N$.

$\exists F: \{0,1\}^* \to \{0,1\}$.

in $TIME_{RAM}(T(n)\log n) \setminus TIME_{RAM}(T(n))$

证: Fix a function T.

$$\text{def } HALT_T(P,x) = \begin{cases} 1, & \text{if } |P| \le \log\log|x|, \text{ 且 } P \text{ halts on } x \text{ in } \le 100T(|P|+|x|) \text{ steps} \\ 0, & \text{other} \end{cases}$$

〈1〉. $HALT_T \in TIME_{RAM}(T(n)\log n)$

1. if $|P| > \log\log|x|$ return 0. $\to O(|P|)$

2. compute $T_0 = T(|P|+|x|)$ $\to O(T(|P|+|x|))$

3. simulate P on x by $100T_0$ steps

$a \cdot |P|^b \cdot 100 T_0$

4. If P halts

return 1

5 else

return 0

total: $O(|P|^b \cdot T_0)$. $|P| \le \log\log|x|$.

$\le O(\log\log(|P|+|x|))$..

$\le O(T_0 \cdot \log\log^b(|P|+|x|))$.

$\le O(T(|P|+|x|) \cdot \log\log^b(|P|+|x|))$

$= O(T(|P|+|x|) \cdot \log(|P|+|x|))$.

(2). $HALT_T \notin TIME_{RAM}(T(n))$.

假设可以

NAND-RAM, $|P| < 0.1 \log\log m$

Q^* = on input z.

1. if z is not of $P1^m$ $O(|P|+m)$.

return 0

2. $b = HALT_T(P, z)$ $O(T(2|P|+m))$

3. if $b=1$,

loop forever

4. else halt

$$Q^*(P1^m) = \begin{cases} \text{loop} & \text{, if } |P| < 0.1\log\log m \text{ 且 } P \text{ halts on } P1^m \text{ within } 100 \cdot \underbrace{T(|P|+|z|)}_{2|P|+m} \quad ① \\ \text{halt} & \quad ② \end{cases}$$

let $P = Q^*$, $m = 2^{2^{100|Q^*|}}$, run $Q^*(Q^* 1^m)$.

total time $2T(2|P|+m)$，为情况 ①. 矛盾！

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定义: $F: \{0,1\}^* \to \{0,1\}$

$F_{\uparrow n}: \{0,1\}^n \to \{0,1\}$, $F_{\uparrow n}(x) = F(x)$.

$F_{\uparrow n} \in Size_n(T(n))$ if need $T(n)$ gates.

$F \in Size(T(n))$ if $\forall n$, $F_{\uparrow n} \in Size_n(T(n))$.

定理: $\forall$ nice function T

$$TIME(T(n)) \subseteq SIZE(T(n)^3)$$

定义: $P/poly = \bigcup_{c=1,2,\dots} SIZE(n^c)$

$P \subsetneq P/poly$

定理: some function in $P/poly$ 不可计算.

证: $S: N \to \{0,1\}^*$.

$n \to$ binary encoding $\to$ remove 最高位 1.

Unary Halting function.

$UH(x) = HALTONZERO(S(|x|))$

UH is uncomputable (归纳即可)

固定长度后 UH 为常数, 可计算

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